

Avni Mohan

avnimohan9@gmail.com | (425)-229-1239
Bellevue, WA
<https://www.linkedin.com/in/avni-mohan-1541461b3>

Education

University of Illinois Urbana-Champaign B.S. Computer Science GPA: 3.73 *Aug. 2021 – Dec. 2024*
Relevant Coursework: Data Structures, Data Science, Discrete Structures, Database Design, Computer Architecture, Algorithms, Numerical Methods, Databases, Distributed Information Systems, Computational Advertising, Experience Design, Social Visualization, Information Security and Privacy, Systems Programming, TCP and IP Networking, UX Theory

Interlake High School GPA: 4.45 Bellevue, WA *Sept. 2017 – Jun. 2021*

Work Experiences

- Machine Learning Research Assistant** | Champaign, IL Jan 2025 – Present
- Analyzing EPICOVID 2.0 data for predicting healthcare utilization in Brazil during and after SARS-CoV-2 infection.
 - Evaluating models including XGBoost, CatBoost, and TADNet using metrics such as AUC, precision, and recall to determine optimal performance
 - Developing a scalable pipeline in Python for cleaning, processing, and training on healthcare data
- Consultant for Illinois Business Consulting** | Champaign, IL September 2024– May 2025
- Utilized Python (Pandas, NumPy) and Excel to analyze segmented enrollment data across multiple client-defined categories.
 - Analyzed U.S. Census and public financial datasets to build a consumer profiles using geodemographic segmentation.
 - Designed data visualizations using Matplotlib and PowerPoint, delivering weekly presentations with strategic insights.
- Business Analyst Intern for American Retail Supply** | Bellevue, WA June 2024– August 2024
- Performed pricing optimization and market analysis using SQL and Python for the development of an e-commerce platform.
 - Used Tableau to create interactive dashboards that visualized sales trends, pricing, and customer segmentation.
 - Integrated a shipping API to automate shipping quote generation, accelerating order processing time by 35%.
- Software Engineering Intern at Click2Cloud** | Bellevue, WA (Remote) June 2023 – August 2023
- Built a GPT-3-based NLP pipeline using OpenAI API and spaCy to extract key concepts and entities from technical reports, reducing manual review time by over 70%.
 - Automated generation of structured PowerPoint summaries using python-pptx with client-specific templates.
- Course Assistant for CS 196 at University of Illinois** | Champaign, IL August 2022 – May 2023
- Mentored honors students in building machine learning-based projects using TensorFlow, OpenAI GPT APIs, and scikit-learn.
 - Reviewed project proposals for feasibility, scalability, and innovation, and developed technical documentation and demos.
- Web Developer at Spurlock Museum** | Urbana, IL January 2022 – May 2023
- Developed front-end features using HTML5, CSS3, JavaScript, and jQuery, integrating with RESTful APIs for dynamic content.
 - Created an interactive museum kiosk interface using React, Node.js, and Bootstrap
- Research Intern at Institute for Systems Biology** | Seattle, WA (Remote) June 2020 – August 2020
- Collaborated with senior research scientists to analyze large-scale transcriptomic datasets using machine learning.
 - Visualized genomic data using Python and MATLAB by applying dimensionality reduction methods such as t-SNE and PCA.
 - Implemented a custom Principal Component Analysis algorithm to identify gene expression patterns and biological clusters.
- Software Engineering Intern at Jalaja** | Puyallup, WA (Remote) June 2018 – July 2019
- Built recommendation engines for an e-commerce platform using collaborative filtering techniques using pandas and NumPy.
 - Developed and maintained e-commerce product pages using Magento, enhancing UI with custom JavaScript widgets.

Skills

- **Languages & Frameworks:** Python (Pandas, NumPy, scikit-learn, Matplotlib, seaborn), Java, C++, SQL, HTML, CSS, JavaScript (Node.js, React, jQuery), MATLAB, R, Verilog, C
- **Machine Learning:** XGBoost, CatBoost, TADNet, TensorFlow, OpenAI API (GPT-3), model evaluation (AUC, precision, recall)
- **Tools & Technologies:** AWS, Google Cloud Platform, Tableau, Excel, Git, Figma, MongoDB, Apache, Magento, Android Studio